

List of Materials and Apparatus for JC2 H2 BIOLOGY Preliminary Examination Paper 4

Question 1

Each candidate will require:

Item	Apparatus / reagents / chemicals	Quantity per student
1	1.0 mol dm ⁻³ sucrose solution in a beaker or container, labelled S1 , provided at room temperature (see Preparation of materials)	70 cm ³
	S2 0.5 mol dm ⁻³ sucrose solution in a beaker or container, labelled S2 , provided at room temperature	70 cm ³
	S3 0.25 mol dm ⁻³ sucrose solution in a beaker or container, labelled S3 , provided at room temperature	at least 100 cm ³
2	Pieces of potato in a beaker or container, labelled P (see Preparation of materials)	5 potato cylinders, each measuring at least 4 cm in length
3	Distilled water in a beaker or container, labelled W , provided at room temperature	at least 200 cm ³
4	[HH] Methylene blue solution, labelled M , provided at room temperature (see Preparation of materials)	at least 15 cm ³
5	Pipettes, plastic or glass with a teat	2
6	Test-tubes – large boiling tubes (to hold more than 25 cm ³ but no more than 50 cm ³)	3
7	Test-tube rack to hold 5 large boiling test-tubes	1
8	Test-tubes – small (capacity 20 cm ³ to 30 cm ³)	4 (with practical 1) should have 12
9	Test-tube rack to hold 5 small test-tubes	1
10	Ruler	1
11	Scalpel or sharp blade	1
12	White tile	1
13	Paper towels	8
14	Forceps	1

Preparation of materials

Solutions

- 1.0 mol dm⁻³ sucrose solution may be prepared the day before the examination. It should be kept in a covered container in a refrigerator.
- 0.5% methylene blue solution may be prepared the day before the examination. It should be kept in a covered container.

The solutions must be at **room temperature** for the examination.

(i) **S1**, 1.0 mol dm⁻³ sucrose solution

This is prepared by sprinkling **68.4 g of sucrose, a little at a time, onto the surface of 80 cm³ of distilled water, stirring continuously as you sprinkle. Make up to 200 cm³ with distilled water.**

- **S2**, 0.5 mol dm⁻³ sucrose solution
100 cm³ of S1 mixed with 100 cm³ of distilled water.
- **S3** 0.25 mol dm⁻³ sucrose solution
100 cm³ of S2 mixed with 100 cm³ with distilled water.
- **[HH] (iii) M**, 0.5% methylene blue solution

This is prepared by putting **0.5 g of methylene blue into 80 cm³ of distilled water** and stirring continuously. Make up to 100 cm³ with distilled water.

(ii) **P**, at least 5 pieces of potato cylinders placed on a damp paper towel in a covered dish, labelled **P**

Preparation of potato cylinders

- You may use any variety of the white (or Irish) potato, *Solanum tuberosum*.
- Use a cork borer to prepare the potato cylinders
Each candidate should be provided with a mixture of different lengths, **varying from 4.0 cm to 4.5 cm.**

The potato pieces for each candidate should be prepared on the day of the examination.

Question 2

The apparatus highlighted in yellow are also used in Practical 1

Item	Apparatus / reagents / chemicals	Quantity per student
1	1.0% glucose solution in a beaker or container, labelled R , provided at room temperature (see Preparation of materials)	at least 50 cm ³
2	0.1% glucose solution in a beaker or container, labelled C1 , provided at room temperature (see Preparation of materials)	at least 20 cm ³
3	Distilled water in a beaker or container, labelled C2 , provided at room temperature (see Preparation of materials)	at least 20 cm ³
4	0.5 mol dm ⁻³ sucrose solution in a beaker or container, labelled C3 , provided at room temperature (see Preparation of materials)	at least 10 cm ³
5	Distilled water in a beaker or container, labelled W , provided at room temperature	at least 100 cm ³
6	Benedict's solution (qualitative) in a beaker or container, labelled Benedict's , provided at room temperature	at least 40 cm ³
7	Beakers or containers, to hold 50 cm ³	5
8	Beaker, capacity approximately 300 cm ³ , with water at 40 °C to 45 °C, suitable for heating as a water-bath. The Supervisor may use a thermostatically controlled water-bath to provide the hot water for candidates. 500 cm ³ beaker	1
9	10 cm ³ syringes	2
10	5 cm ³ syringe	1
11	Test tubes small, capacity 20 cm ³ to 30 cm ³ , suitable for heating	8 + 4 (practical 2) = 12
12	Test-tube holder, to hold hot test-tubes	1
13	Glass rod	1

Item	Apparatus / reagents / chemicals	Quantity per student
14	Bunsen burner, bench mat, gauze and tripod to support water-bath	1
15	Wooden rack to hold boiling tubes and test tubes	1
16	Thermometer	1
17	Container, capacity approximately 200 cm ³ , with tap water, labelled For washing	1
18	Container (capacity approximately 200 cm ³), labelled For waste	
19	Stopwatch	1
20	Safety goggles	1 pair
21	Paper towels	8
22	Glass marker pen (permanent)	1
23	Access to water from thermostatically-controlled water bath set at 40 to 45 °C	-

Preparation of materials

1. R, 1.0% glucose solution

This is prepared by sprinkling 1.0 g of glucose into 80 cm³ of warm distilled water and mixing well. Make up to 100 cm³ with distilled water.

Solutions **C1** is made using 1.0% glucose solution. This is prepared as described in (i) for **R**.

2. C1, 0.1% glucose solution

This is prepared by putting 10 cm³ of 1.0% glucose solution into 90 cm³ of distilled water and mixing well.

3. C2, distilled water

This is prepared by putting distilled water

4. C3, 0.5 mol dm⁻³ sucrose solution

See preparation on practical 2.

Question 3

- Each candidate must have **sole, uninterrupted** use of the prepared slide for 30 **minutes** only.

Item	Apparatus / reagents / chemicals	Quantity per student
1	Microscope For each candidate: • microscope must be set up on low power • slide must not be left on the stage of the microscope.	1
2	Prepared slide, labelled P1 (placed in a Petri dish)	1

P1: slides with Human Lymph gland